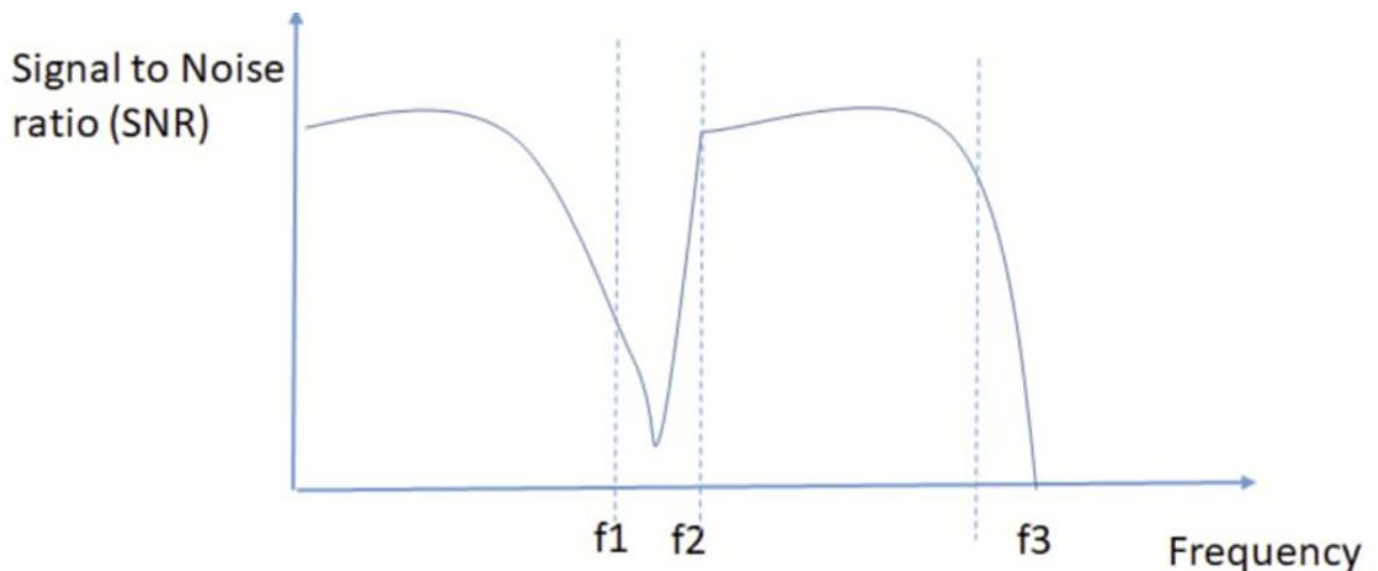
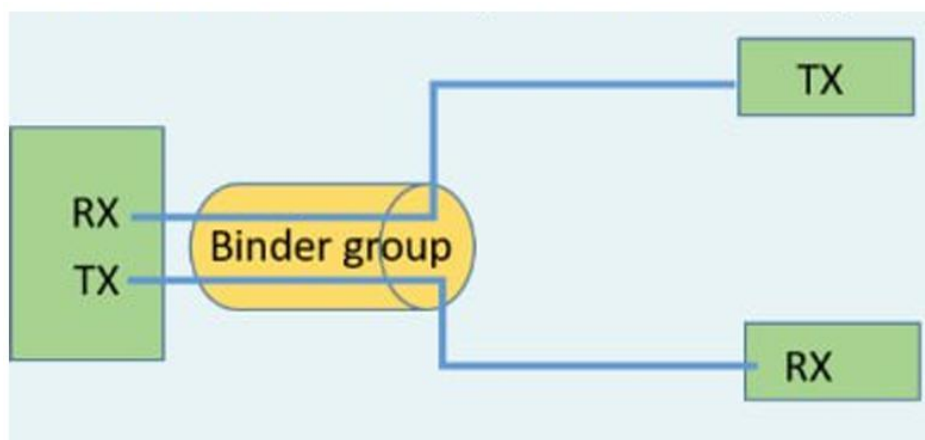


1. The hidden terminal problem in Wi-Fi is mainly due to:
 - a. a. The high signal power that is transmitted by an access point compared to that of a mobile host.
 - b. b. The use of unlicensed spectrum bandwidth.
 - c. c. The high mobility of the wireless nodes.
 - d. d. The attenuation of the signal in the wireless channel that makes some nodes not in the radio range of others.
2. Which one of these FTTX configurations is the best in terms of expected bit rate?
 - a. a. FTTB - to the building
 - b. b. FTTC - to the curb
 - c. c. Fiber optics in the VDSL+
 - d. d. FTTE - to the exchange
3. Which statement correctly describes the function of a Base Station (or eNodeB) in a cellular network compared to an Access Point (AP) in a Wi-Fi network?
 - a. a. Both are primarily used for peer-to-peer communication between mobile devices.
 - b. b. Both typically manage user mobility and handovers between different cells/APs equally.
 - c. c. An Access Point uses licensed frequency spectrum, while a Base Station operates in unlicensed bands.
 - d. d. A Base Station covers a much larger geographic area and is part of a standardized, centrally coordinated wide-area network infrastructure.
4. Which of the following is NOT a typical characteristic of a wireless link compared to a wired link?
 - a. a. Constant and predictable signal propagation without reflections
 - b. b. Decreased signal strength during the signal propagation through the transmission medium
 - c. c. Interference from other devices operating in the same frequency bands
5. Passive Optical Networks benefits What are the benefits of Passive Optical Networks (PON) compared to Active Optical Networks?
 - a. The infrastructure cost is lower due to simpler equipment and fewer active components.
 - b. It's easier to manage thanks to the use of Ethernet-based approaches.
 - c. You avoid the need for Optical Network Units at the user side.
 - d. You only need to replace damaged components, not the entire system.



6. ADSL channel behavior and Discrete Multitone Technology (DMT) Considering an ADSL line with frequencies fully attenuated (low SNR) in certain ranges, how should a modulation based on Discrete Multitone Technology [DMT] handle these frequencies?
- It should include transmitting "tones" in the frequency band below f_1 and between f_2 and f_3 .
 - It must include only transmitting "tones" in the frequency band between f_2 and f_3 .
 - It should include transmitting "tones" only in the frequency band below f_1 .



7. Cross-talk in shared cable paths What type of cross-talk might arise when two pairs of cables share an initial path before diverging?
- The Near End Cross Talk.
 - None of the cross talks may arise in this setting.
 - Both the two kinds of cross-talks may arise.
 - The Far End Cross Talk.
8. FTTX configurations and deployment cost Which of the FTTX configurations has a higher deployment cost?
- FTTB [] to the building.
 - The cost is identical, given the fact that fibers shall be deployed in this access network part.
 - FTTC [] to the curb.

d. FTTE π to the exchange.

9. Water filling algorithm in ADSL When are the advantages of applying the water filling algorithm in an ADSL system most noticeable?
- The channel behavior both in the upstream and downstream has a flat (constant) signal-to-noise ratio.
 - It is important to separate the telephone bandwidth from the one used by the data.
 - It is important to split the transmission into two parts: downstream and upstream.
 - The channel behavior both in the upstream and downstream has not a flat (constant) signal-to-noise ratio.
10. VDSL improvements Which are the two improvements enabling the VDSL?
- The use of the DMT modulation in the copper cables and the full cancellation of the cross-talk noise from the CO to the end user.
 - The use of high frequencies in the fiber optic cables and the cancellation of the interference arising in the fiber optic domain.
 - The vectoring of the signals in the fibers cables that arrive till the end users and the positioning of the ONU close to the end users.
 - The vectoring of the signals in the copper cables and the placement of the DSLAM (where the fiber ends) close to the end user.
11. Importance of latency in future wireless systems. In the future generation wireless systems, the latency will become a key performance metric. Why is it becoming so important?
- Future mobile applications will use high data processing and computation.
 - Many mobile applications will require high responsiveness in the interaction with the network.
 - The end users may have high mobility during their connection.
 - Many mobile applications will require high throughput (bit rate) at the end users.
12. Critical impairment in wireless infrastructure Which is the most critical impairment in a wireless infrastructure when multiple end users access the wireless link?
- The multipath, since we have multiple users.
 - The path loss, since multiple users may be very distant from the base station.
 - The interference that they may generate one with the other if the multiple access is not suitably controlled.
13. Water filling algorithm in ADSL Suppose to apply the water filling algorithm in an ADSL system. The main advantages of this approach are attained when:
- The channel behaviour both in the upstream and in the downstream has a flat (constant) signal to noise ratio.
 - It is important to separate the telephone bandwidth for the one used by the data.
 - It is important to split in two parts the transmission in the downstream and in the upstream.

- d. The channel behaviour both in the upstream and in the downstream has not a flat (constant) signal to noise ratio.

14. Best configuration in terms of expected bit rate Which one of these configurations is the best in terms of expected bit rate?

- a. FTTC to the curb.
- b. ADSL.
- c. FTTE to the exchange.
- d. FTTB to the building.

15. Role of the EDGE in a network What is the EDGE of a network and which is its role?

- a. The very core part of a network where powerful network functions are implemented.
- b. That part of the network between the access and the core with some intelligent functions.
- c. The very peripheral part of the network where only simple functions are implemented.
- d. The very peripheral part of the network where all the applications are.

16. Cross-talk noise on ADSL What is the cross-talk noise on ADSL?

- a. A noise that is generated at a generic receiver within a copper cable.
- b. The coupling of signals from a circuit to another giving rise to undesired interference.
- c. The interference of a signal in a frequency band to signals in other frequency bands of the ADSL.
- d. A noise that is generated at a generic receiver within a fiber cable.

17. Evolution of the access network The evolution of the current access network aims at:

- a. Employing fiber cables instead of copper ones.
- b. Providing an ADSL access to all the users.
- c. Using in a more capillary way the already existing infrastructure based on copper cables.
- d. Using only wireless access given its flexibility and low cost. Question

18. Key element of ADSL success One of the key elements of ADSL success is related to:

- a. The use of the same copper cable of the old telephone network.
- b. The fact that no new network elements are needed at the Central Office side.
- c. The use of the same frequency band of the old telephone network.
- d. The use of new fiber cables in place of the old telephone ones.